

Task 2: Exploratory Data Analysis & Business Intelligence

1.TOOLS USED:

PYTHON | PANDAS | NUMPY |
MATPLOTLIB | SEABORN | POSTGRESQL

2.DATASET: SALES DATASET

3.SUBMITTED BY: DEEPAK KUMAR

Objectives of the Analysis

1. Understand the structure and characteristics of the sales dataset.
2. Perform descriptive statistics and exploratory data analysis.
3. Analyze numerical and categorical variables.
4. Identify important sales patterns and distributions.
5. Answer business-oriented questions using SQL.
6. Analyze relationships between numerical variables.
7. Extract meaningful business insights from the data.

Data Preparation

- The cleaned sales dataset was loaded into Python using Pandas.
- **Preparation Steps**
- Loaded the cleaned CSV file.
- Verified the dataset shape.
- Checked column names and data types.
- Reviewed numerical and categorical variables.
- Prepared the dataset for statistical analysis and visualization.
- **Dataset Shape**

Rows: 1,000

Columns: 16

Descriptive Statistics

Variable	Mean	Minimum	Maximum
Age	41.35	18	65
Quantity	5.44	1	10
Unit Price	₹25,486.78	₹145.78	₹49,997.53
Total Sales	₹139,399.44	₹437.34	₹493,677.50

- Descriptive statistics were used to understand the central tendency and variation of numerical variables.
- The median Total Sales is **₹108,594.03**, while the standard deviation is **₹114,100.05**, showing substantial variation in sales values.

Categorical Analysis

Gender Distribution

- Male: 511 records
- Female: 489 records

Product Distribution

- Mobile: 184
- Book: 178
- Laptop: 170
- Chair: 159
- Shoes: 156
- Rice: 153

Category Distribution

- Electronics: 354
- Education: 178
- Furniture: 159
- Fashion: 156
- Grocery: 15

City Distribution

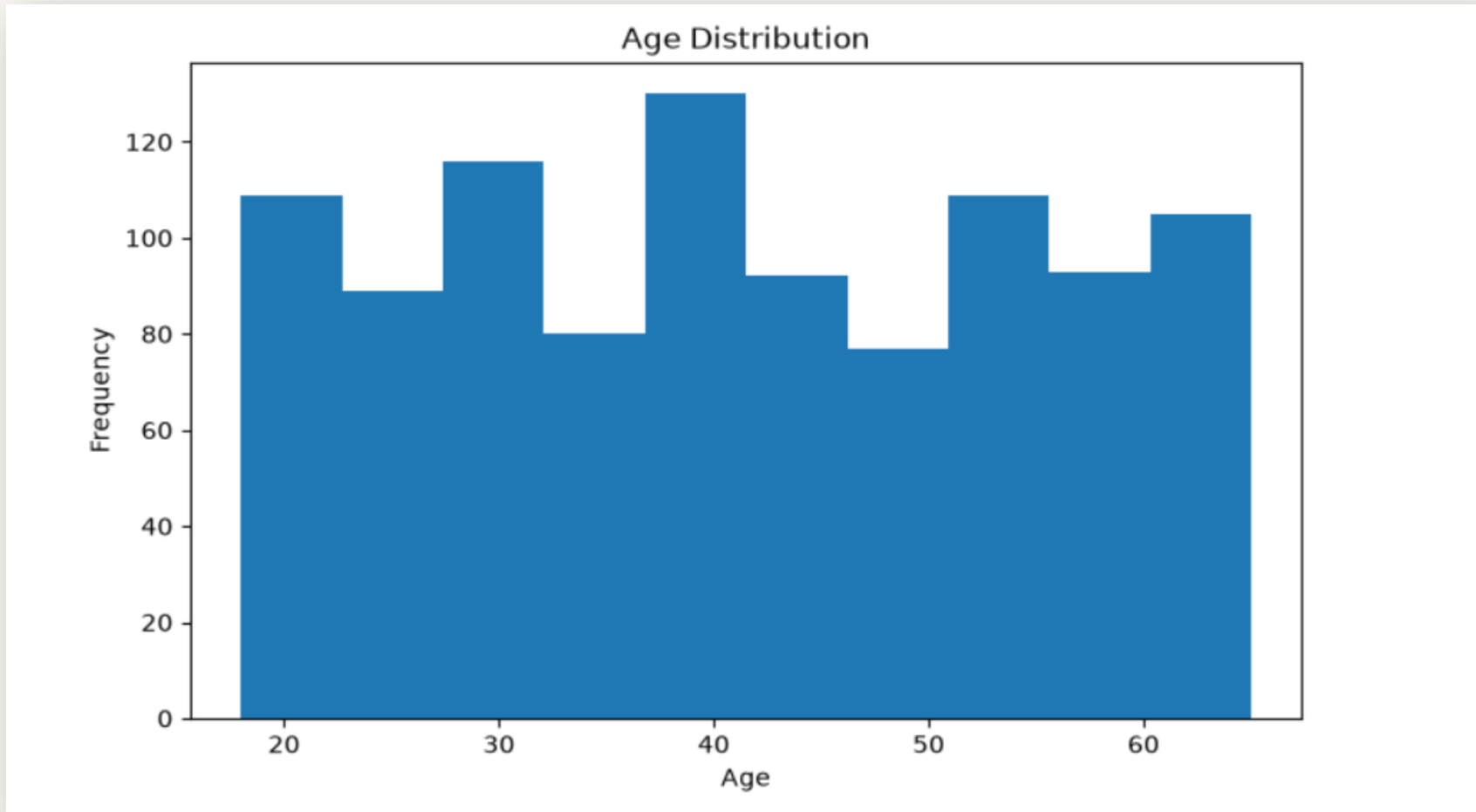
- Patna: 148
- Kolkata: 133
- Mumbai: 131
- Hyderabad: 125
- Delhi: 125
- Bengaluru: 122
- Gaya: 117
- Pune: 99

Monthly Distribution

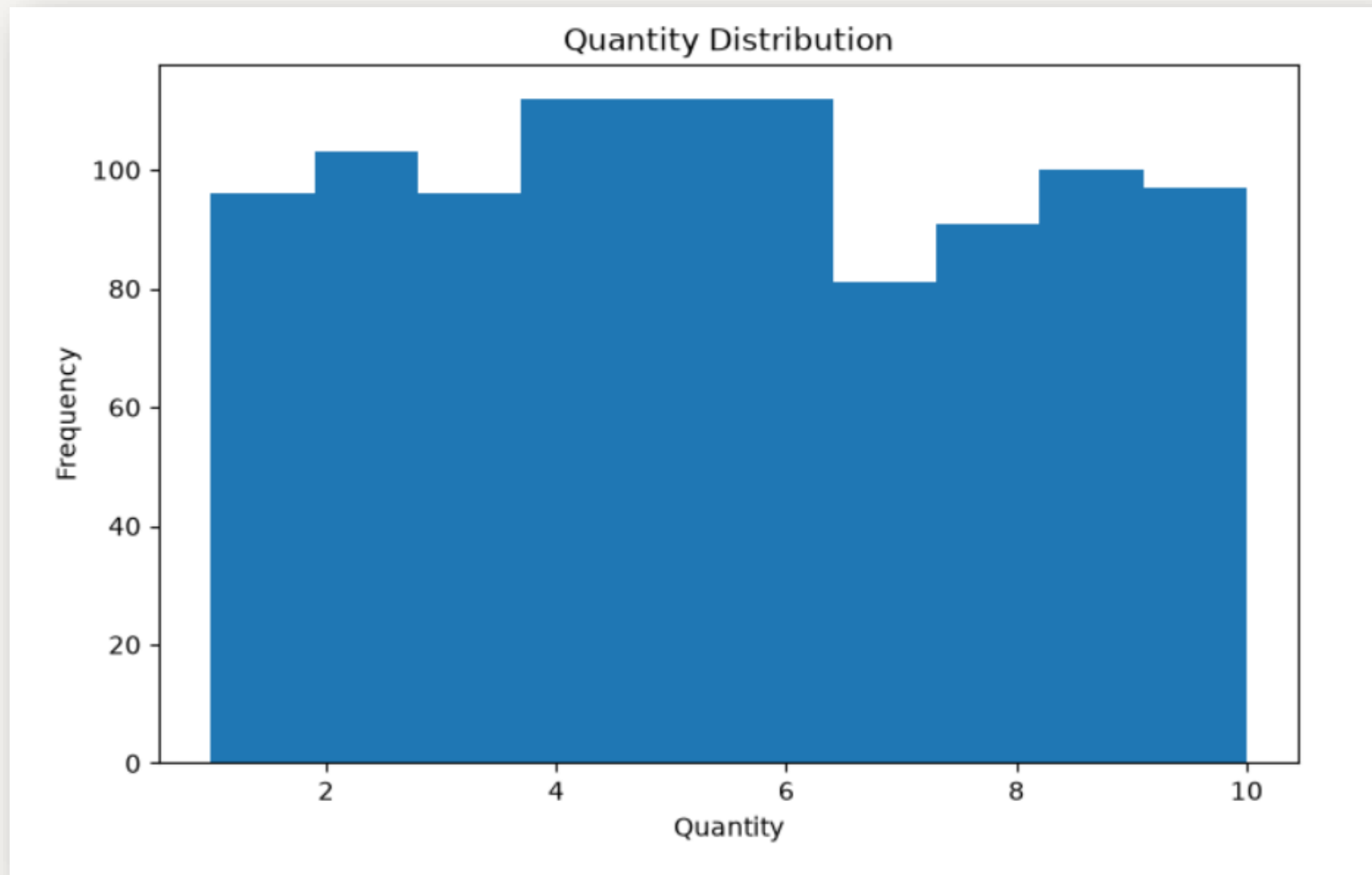
- December: 98
- March: 89
- February: 86
- May: 85
- October: 84
- July: 83
- June: 83
- January: 81
- April: 80
- September: 79
- November: 77
- August: 75

December has the highest number of records, while August has the lowest.

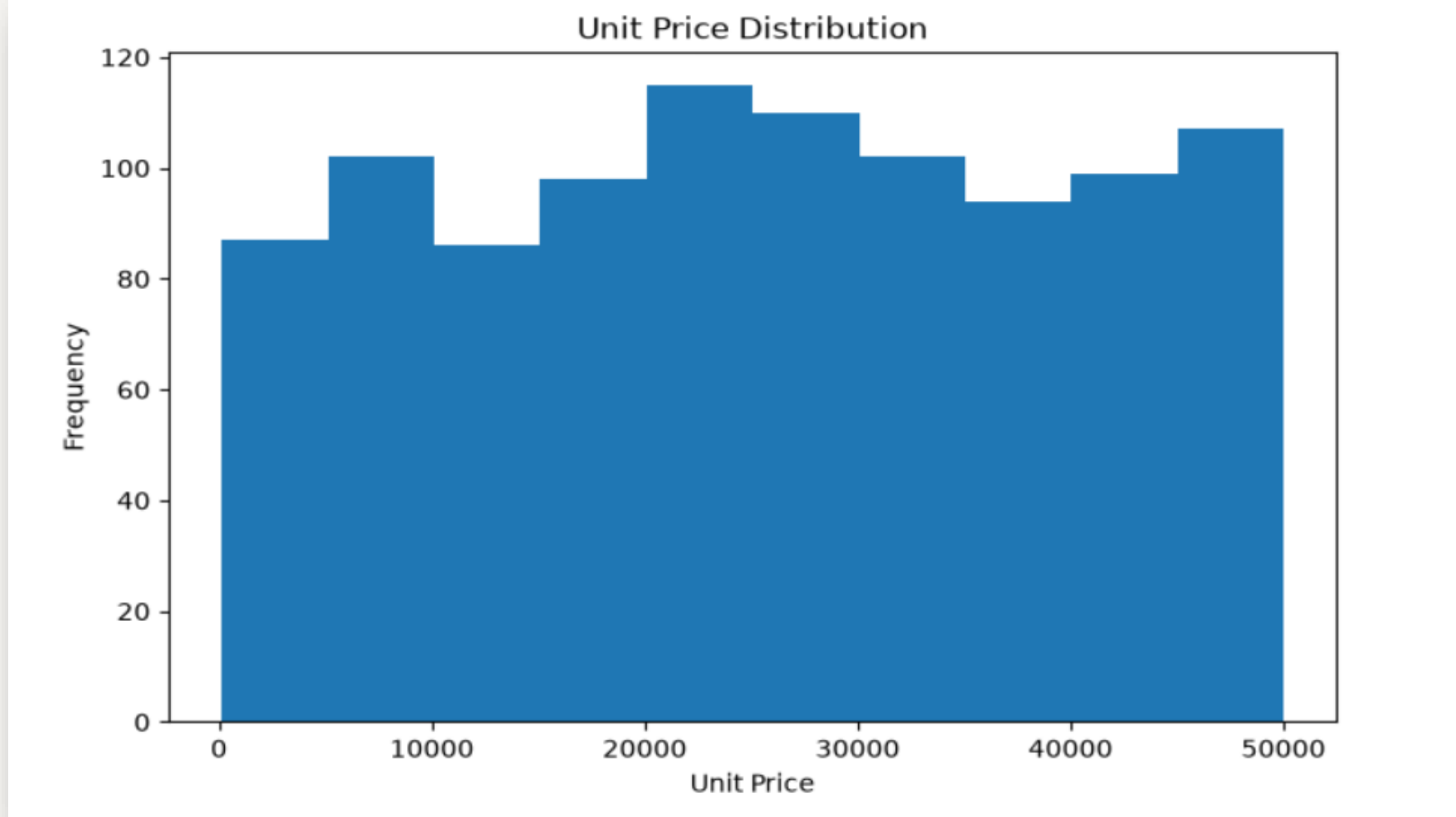
Visualizations Created



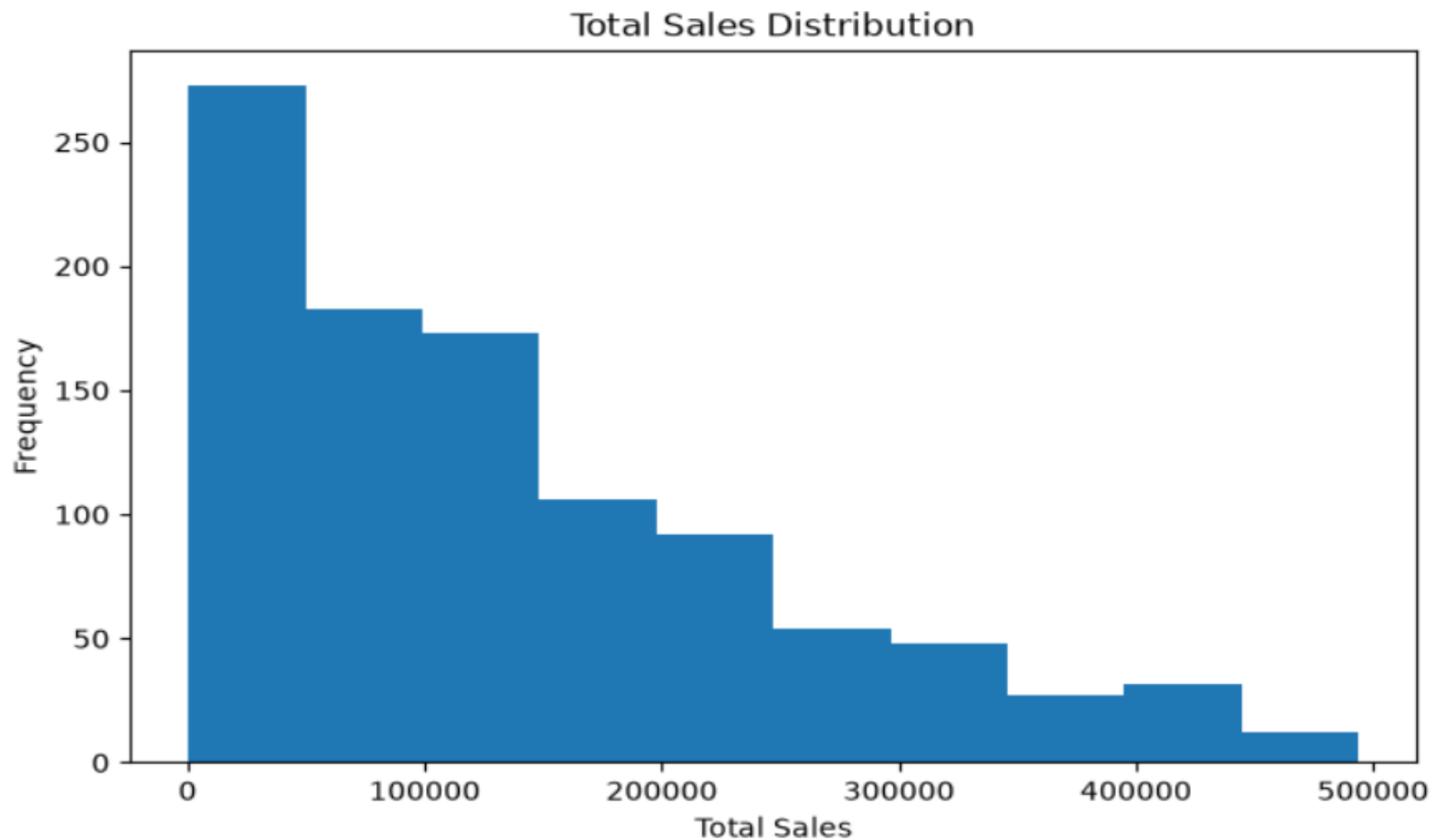
- Customers are distributed across a broad age range, approximately **18 to 65 years**.
- The distribution is relatively balanced, with no single age group dominating the dataset.
- This indicates that the business serves customers from different age segments.



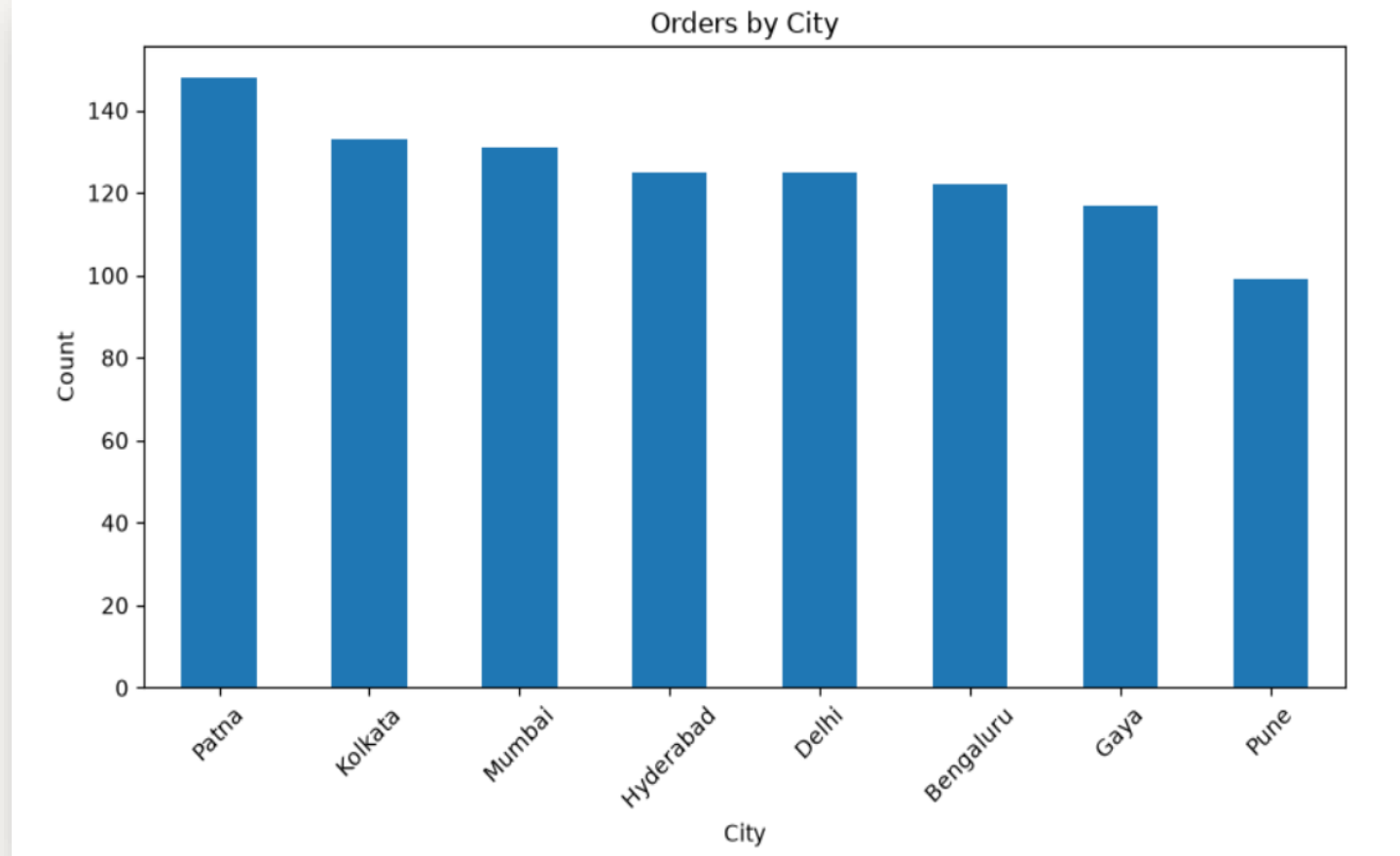
- The quantity purchased ranges from approximately **1 to 10 units per order**.
- Most quantity levels have relatively similar frequencies.
- Quantities around **4–6 units** appear comparatively frequent.



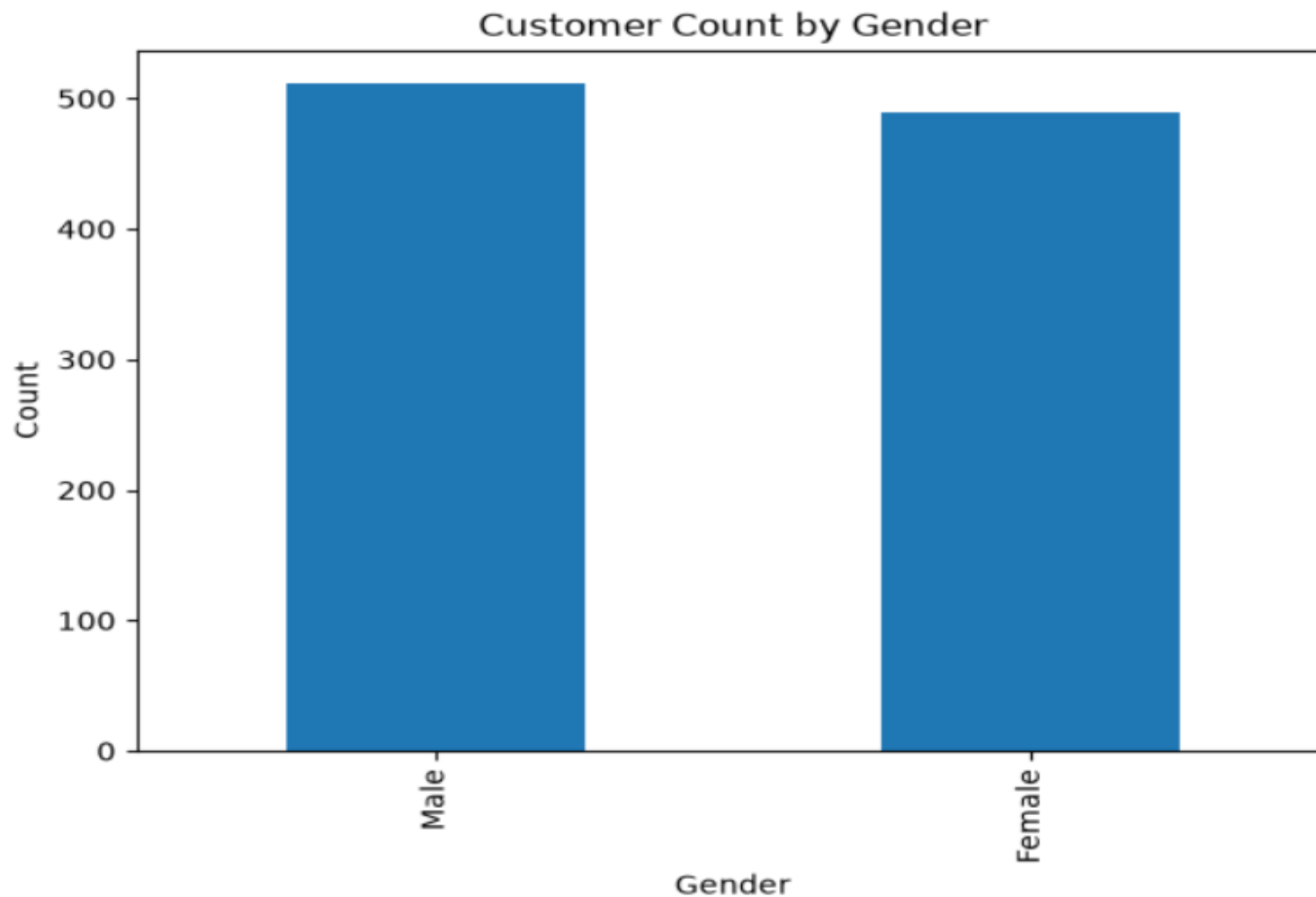
- Unit prices range approximately from **0 to 50,000**.
- The distribution is fairly spread across the price range.
- A significant number of products fall in the **20,000–30,000** price range.
- This suggests the dataset contains products across multiple price segments.



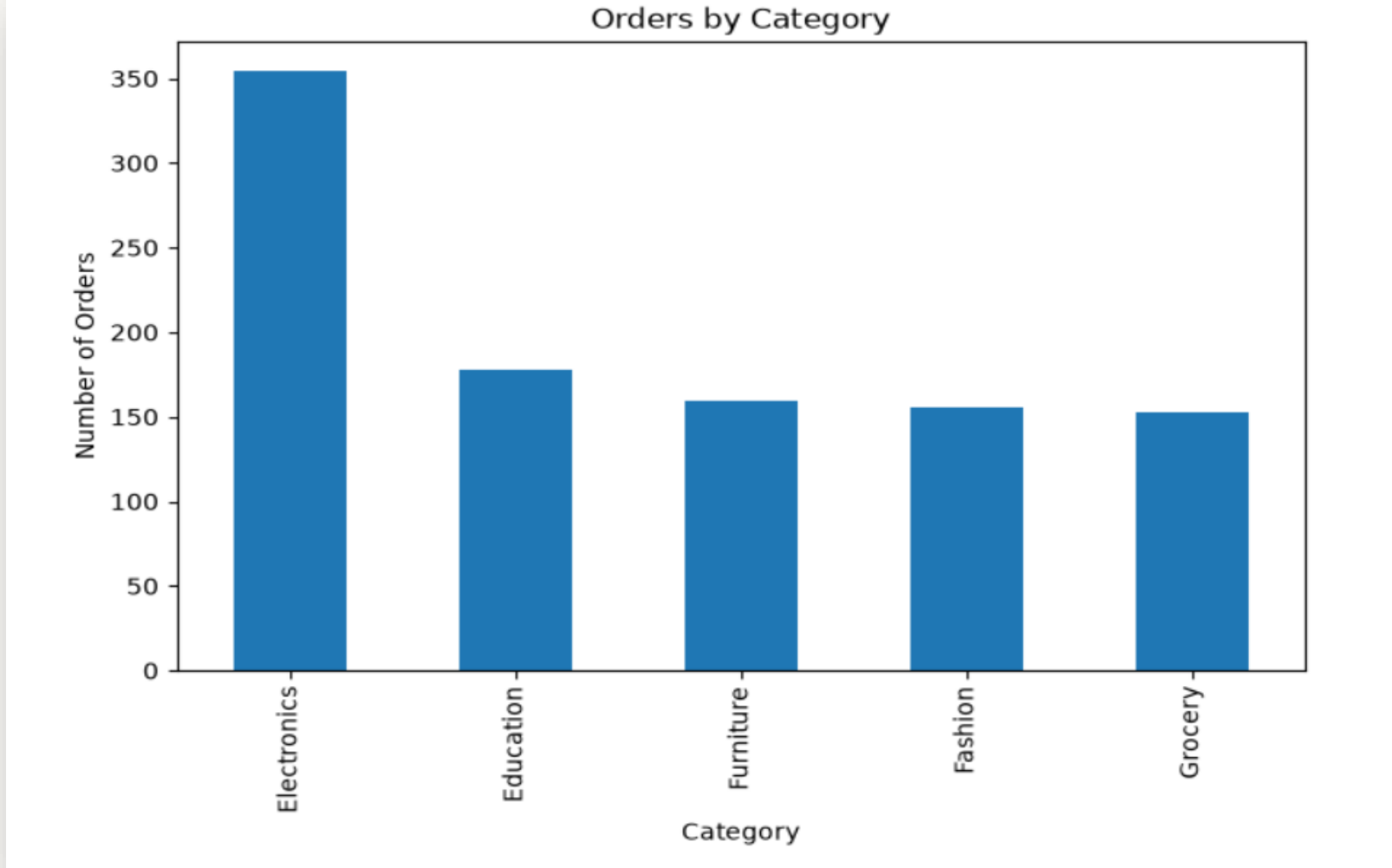
- Total sales values range approximately from **0 to 500,000**.
- The distribution is **right-skewed**.
- A large number of transactions have relatively lower sales values.
- Only a smaller number of transactions generate very high sales values.



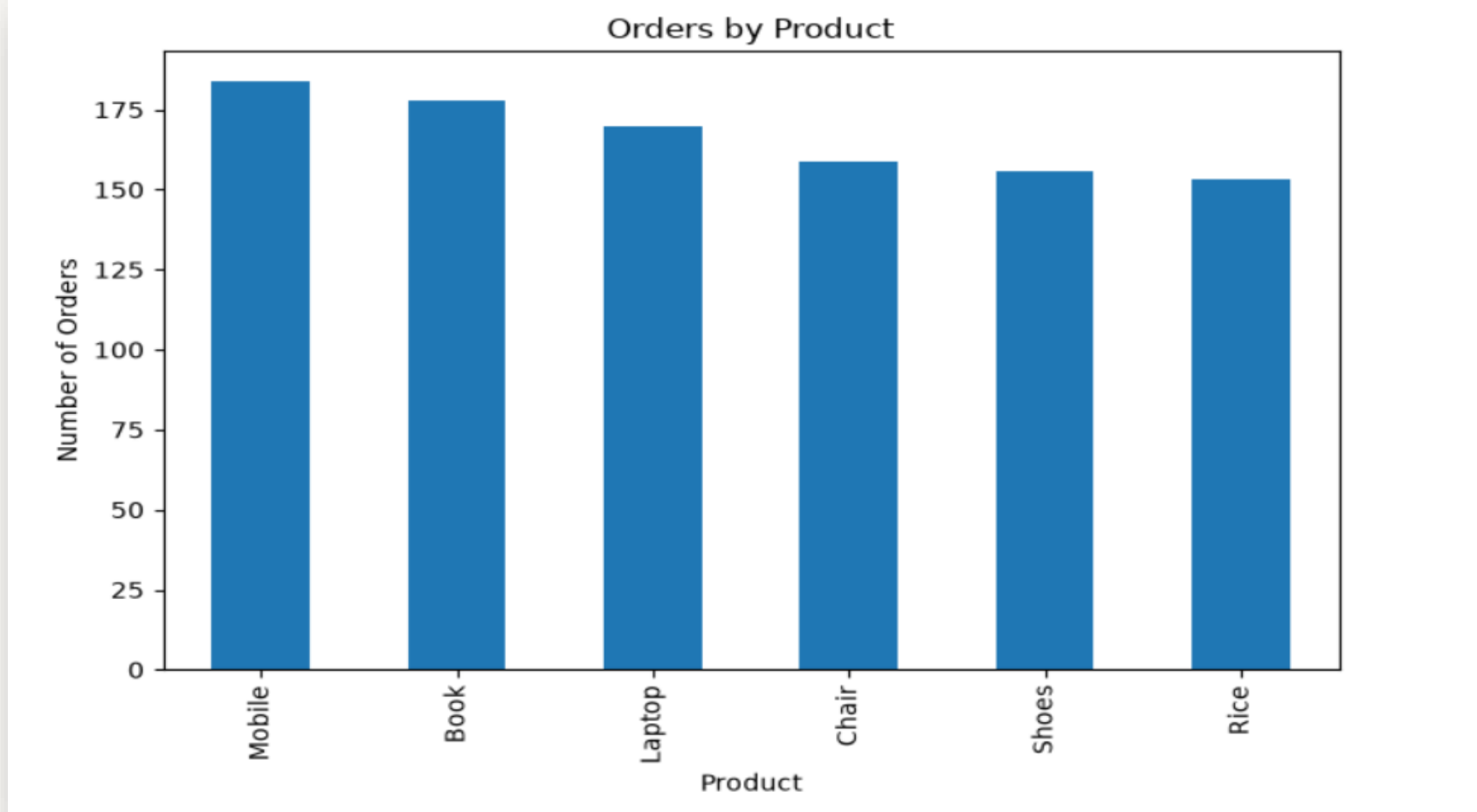
- **Patna** has the highest number of orders, with approximately **148 orders**.
- Kolkata and Mumbai follow with around **130+ orders** each.
- Hyderabad and Delhi have approximately **125 orders**.
- Pune has the lowest order count among the displayed cities, at around **100 orders**.
- Overall, order distribution across cities is relatively balanced.



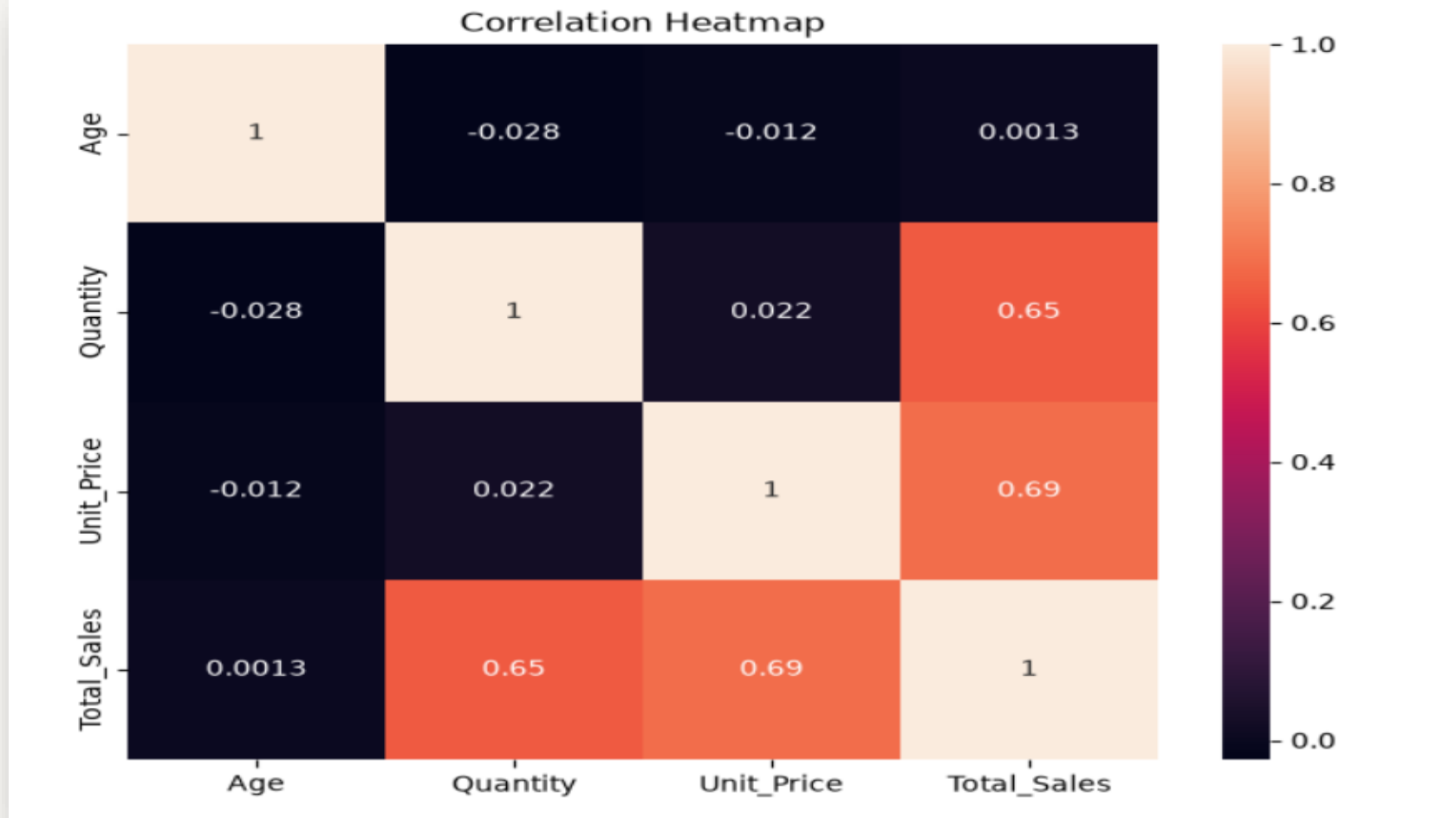
- The dataset contains approximately **510 male** customers/orders and **490 female** customers/orders.
- The difference between male and female representation is relatively small.
- This indicates a **nearly balanced gender distribution**.



- **Electronics** is clearly the leading category with approximately **355 orders**.
- Education is the second-highest category with around **175–180 orders**.
- Furniture, Fashion, and Grocery have comparatively fewer orders.
- Electronics has substantially higher demand than the other categories.



- **Mobile** has the highest number of orders, approximately **185**.
- **Book** follows with around **178 orders**.
- **Laptop** has approximately **170 orders**.
- Chair, Shoes, and Rice have slightly lower order counts.
- Mobile appears to be the most frequently ordered product.



- Unit Price and Total Sales = 0.69 → strong positive relationship.
- Quantity and Total Sales = 0.65 → strong positive relationship.
- Age has almost no correlation with Total Sales (0.0013).
- Quantity and Unit Price are almost independent (0.022).
- Therefore, sales are more strongly associated with price and quantity than with customer age.

Overall Business Insights

- **Electronics is the leading category** by order volume, showing strong customer demand.
- **Mobile is the most frequently ordered product**, followed by Books and Laptops.
- **Patna records the highest order volume** among the analyzed cities.
- **Total Sales is strongly associated with Unit Price and Quantity**, with correlations of 0.69 and 0.65 respectively.
- **Age has almost no relationship with Total Sales**, suggesting that purchasing value is driven more by transaction-level factors than customer age.

SQL Business Analysis

- **Top 5 Products by** Identify the highest revenue-generating products in the **latest 6 months**.

Rank	Product	Revenue
1	Laptop	13,009,494.74
2	Book	12,739,832.53
3	Chair	11,560,952.45
4	Shoes	11,149,392.03
5	Rice	10,215,563.97

- **Laptop generated the highest revenue** among the top 5 products.
- Book and Chair also contributed significantly to revenue.
- The difference between the top and fifth product shows a relatively competitive product mix.
- These products can be prioritized for inventory availability and promotional campaigns.

Category-wise Revenue

Rank	Category	Total Revenue
1	Electronics	50,778,581.70
2	Education	25,031,689.40
3	Grocery	22,231,711.28
4	Furniture	21,521,561.48
5	Fashion	19,835,895.79

- **Electronics is the clear revenue leader**, generating more than **₹50.7 million**.
- Electronics revenue is approximately **2× the Education category**.
- Fashion has the lowest revenue among the five categories.
- The business can investigate why Electronics performs strongly and identify opportunities to improve lower-performing categories.

City-wise Revenue

Rank	City	Revenue
1	Patna	20,826,584.43
2	Kolkata	18,884,349.57
3	Bengaluru	18,773,574.32
4	Mumbai	~18M+

- **Patna generated the highest revenue** among the analyzed cities.
- Kolkata and Bengaluru also show strong revenue performance.
- The relatively close performance of the next cities indicates that revenue is not concentrated in only one market.
- High-performing cities can be targeted for stronger inventory and marketing activities.

Monthly-wise Revenue

- January 2025 revenue was approximately ₹10.10 million.
- February 2025 increased to approximately ₹11.51 million.
- March 2025 further increased to approximately ₹13.06 million.
- October 2025 revenue was approximately ₹12.50 million.
- November 2025 increased to approximately ₹12.63 million.
- December 2025 recorded approximately ₹12.30 million.
- January 2026 dropped to approximately ₹0.81 million.

Gender-wise Revenue

Gender	Orders	Revenue
Male	511	72,463,550.63
Female	489	66,935,889.02

- Male customers generated higher revenue than female customers.
- Male customers contributed approximately **₹72.46 million** in revenue.
- Female customers contributed approximately **₹66.94 million**.
- Order counts are almost evenly distributed: **511 male vs 489 female**.
- This suggests that the revenue difference is not mainly due to a large difference in customer count.

Overall Business Insights

- Electronics is the strongest revenue-generating category.
- Laptop is the leading product by revenue in the analyzed top-5 period.
- Patna is the highest-revenue city in the city-wise analysis.
- Customer orders are almost evenly split between male and female customers.
- Male customers generated slightly higher overall revenue.
- Average Order Value is ₹140,523.63.
- Mobile has the highest quantity sold, while Laptop leads in revenue among the visible product results.
- Monthly revenue shows significant variation, highlighting the need for better trend monitoring.